

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)

Total Marks: 25

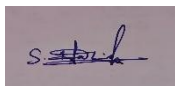
Question Count: 5

Code of Conduct

I S. Harish (192511119) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems

- Examine IoT-cloud architecture, benefits, challenges, and real-world applications.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

Topic:**Introduction:**

- IoT devices generate large volumes of real-time data, while cloud platforms provide the infrastructure needed to store and process this data efficiently.
- IoT-cloud systems enable automation, real-time analytics, intelligent decision-making, and improved resource utilization.
- This integration forms the foundation of connected ecosystems used in smart homes, healthcare, manufacturing, agriculture, transportation, and smart cities.
- The combination of IoT and cloud computing is therefore an important technology for building scalable, intelligent, and connected solutions.

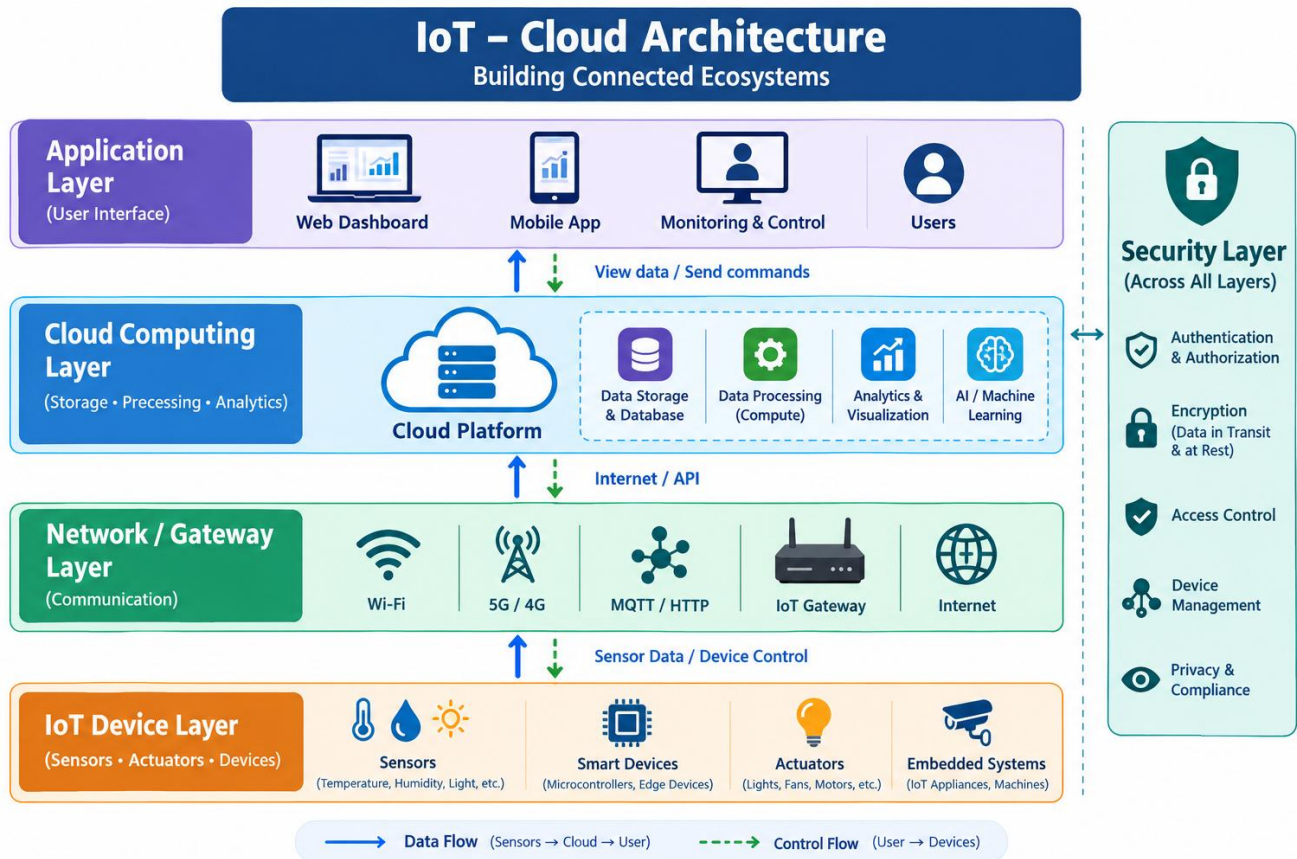
Objective:

- To analyse the benefits of integrating IoT with cloud computing, including scalability, flexibility, and cost efficiency.
- To identify key challenges and security concerns in IoT-cloud environments.
- To explore real-world applications of IoT and cloud computing in smart homes, healthcare, industries, agriculture, and smart cities.
- To understand how IoT-cloud integration supports automation, real-time monitoring, and intelligent decision-making.

Core technical explanation:

- IoT devices generate huge amounts of real-time data, creating challenges in data storage, processing, and management.
- Traditional IoT systems may have limited computing and storage capabilities, making large-scale deployment difficult.
- Data transferred between IoT devices and cloud platforms can introduce latency, security, privacy, and network dependency issues.
- Therefore, there is a need for an efficient IoT-cloud architecture that can securely connect devices, process data, and provide scalable cloud resources.

Architecture or process diagram:



Real-world use case:

1. Smart Homes

- IoT sensors collect data on temperature, motion, lighting, and energy usage.
- Cloud platforms store and analyse the data.
- Users can remotely monitor and control lights, fans, security systems, and appliances.

2. Smart Healthcare

- Wearable devices continuously monitor heart rate, blood pressure, oxygen level, and other parameters.
- Data is securely transferred to cloud platforms for storage and analysis.
- Doctors can remotely monitor patients and receive alerts for abnormal conditions.

3. Smart Agriculture

- Sensors monitor soil moisture, temperature, humidity, and crop conditions.
- Cloud platforms analyze the collected data and provide irrigation recommendations.

- Farmers can remotely monitor fields and automate irrigation systems.

4. Smart Cities

- IoT devices monitor traffic, parking, street lighting, waste management, and environmental conditions.
- Cloud computing processes data from multiple locations.
- Authorities can use real-time information to improve urban services and resource management.

5. Industrial IoT (IIoT)

- Machines and sensors continuously send operational data to the cloud.
- Cloud analytics can detect equipment abnormalities and predict potential failures.
- Enables predictive maintenance, reduced downtime, and improved productivity.

6. Connected Transportation

- Vehicles and transportation systems generate data about location, traffic, fuel consumption, and vehicle condition.
- Cloud platforms process this data for route optimization and fleet management.
- Supports real-time tracking and intelligent transportation services.

Conclusion and key takeaway:

- IoT enables real-time data collection, while cloud computing provides the required storage, processing, and analytics capabilities.
- The proposed IoT-cloud architecture addresses key challenges such as scalability, centralized management, real-time monitoring, and remote access.
- This integration supports practical applications in smart homes, healthcare, agriculture, industries, transportation, and smart cities.
- Overall, IoT-cloud integration enables automation, intelligent decision-making, efficient resource utilization, and better-connected services.

Photo of the presentation:

